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Metagenomic insights into the antibiotic resistome of mangrove sediments and their association to socioeconomic status

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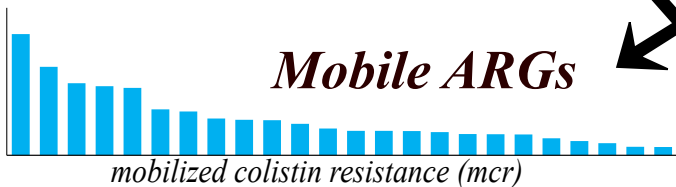
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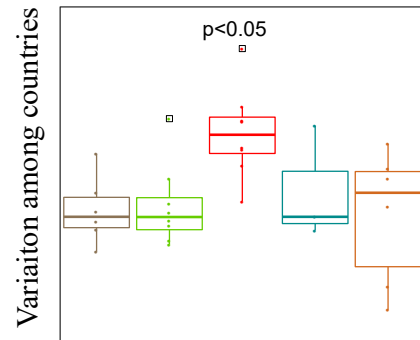
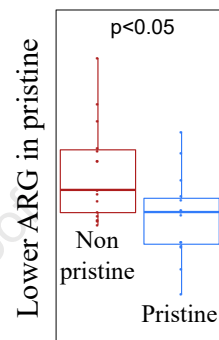
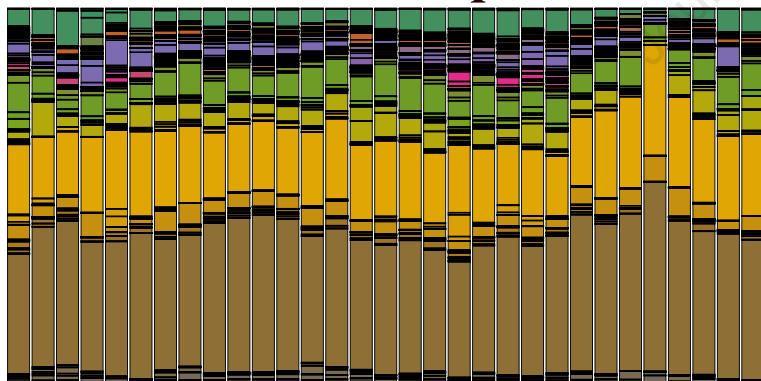
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Factors determining ARG diversity

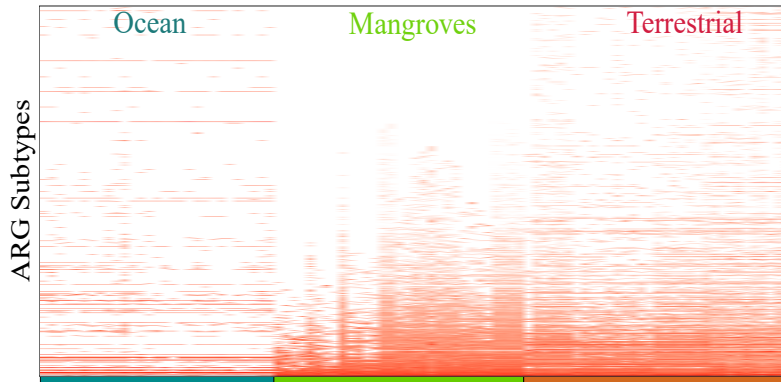
Global mangrove sediments
 ↓
 Whole metagenome sequence
 ↓
 ARG Resistome (Machine learning)



ARG composition



Comparison to other ecosystems



Metagenomic insights into the antibiotic resistome of mangrove sediments and their association to socioeconomic status

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ABSTRACT

Mangrove sediments are prone to anthropogenic activities that could enrich antibiotics resistance genes (ARGs). The emergence and dissemination of ARGs are of serious concern to public health worldwide. Therefore, a comprehensive resistome analysis of global mangrove sediment is of paramount importance. In this study, we have implemented a deep machine learning approach to analyze the resistome of mangrove sediments from Brazil, China, Saudi Arabia, India, and Malaysia. Geography ($R_{ANOSIM}=39.26\%$; $p<0.005$) as well as human intervention ($R_{ANOSIM}=16.92\%$; $p<0.005$) influenced the ARG diversity. ARG diversity was also inversely correlated to the human development index (HDI) of the host country ($R=-0.53$; $p<0.05$) rather than antibiotics consumption ($p>0.05$). Several genes including multidrug efflux pumps were significantly ($p<0.05$) enriched in the sites with human intervention. Resistome was consistently dominated by *rpoB2* ($19.26\pm0.01\%$), multidrug ABC transporter ($10.40\pm0.23\%$), *macB* ($8.84\pm0.36\%$), *tetA* ($4.13\pm0.35\%$), *mexF* ($3.26\pm0.19\%$), *CpxR* ($2.93\pm0.2\%$), *bcrA* ($2.38\pm0.24\%$), *acrB* ($2.37\pm0.18\%$), *mexW* ($2.19\pm0.17\%$), and *vanR* ($1.99\pm0.11\%$). Besides, mobile ARGs such as *vanA*, *tet(48)*, *mcr*, and *tetX* were also detected in the mangrove sediments. Comparative analysis against terrestrial and ocean resistomes showed that the ocean ecosystem harbored the lowest ARG diversity ($S_{chao1}=71.12$) followed by mangroves ($S_{chao1}=258.07$) and terrestrial ecosystem ($S_{chao1}=294.07$). ARG subtypes such as *abeS* and *qacG* were detected exclusively in ocean datasets. Likewise, *rpoB2*, multidrug ABC transporter, and *macB*, detected in mangrove and terrestrial datasets, were not detected in the ocean datasets. This study shows that the socioeconomic factors strongly determine the antibiotic resistome in the mangrove. Direct anthropogenic intervention in the mangrove environment also enriches antibiotic resistome.

Keywords: Mangrove Sediments; Antibiotic Resistance Genes; Resistome; Whole

Metagenome; Machine Learning; Multidrug Resistance

In this study, we show that antibiotic resistance gene diversity in global mangrove sediments is associated to anthropogenic activities as well as socioeconomic status of the country

INTRODUCTION

Antibiotic resistance is the ability to resist the inhibitory effect(s) of an antibiotic. Such resistance has made the treatment of bacterial infections extremely difficult. Over 700,000 deaths related to antibiotic resistance are reported globally which is projected to rise sharply in the coming decades (Friedrich, 2019). Antibiotic resistance has become an acute global concern. For instance, the mortality due to methicillin-resistant *Staphylococcus aureus* (MRSA) alone is comparable to that of HIV (Coates et al., 2011). Multidrug resistant bacteria are not only confined to the hospital environments but also in environmental soil, wastewater treatment plants, gut microbiome, etc. (Diaz et al., 2017). The prevalence of ARGs in the environment could lead to the dissemination of the clinically relevant ARGs to the surroundings and distant locations (Zeng et al., 2019). Rapid and exponential dissemination of ARGs is not only confined at the local or national level but also spreads on a global scale. Although antibiotic resistance is a global issue, the consequences of antibiotic resistance are more severe in developing countries (Klein et al., 2018). Infections caused by multidrug-resistant (MDR) microbes are also associated with higher morbidity, mortality, and healthcare costs (Lee et al., 2007). Hence, infections by MDR pathogens affect the nation's economy as well as the patient's family due to the complications in treatments, costly drugs, and the occupancy of the hospital (Malik and Bhattacharyya, 2019).

The coastal environment such as a saline tolerant mangrove ecosystem serves crucial biogeochemical cycles and home to a myriad of flora and fauna. However, they are also the receiving end of waste inputs from sewage and other sources. Hence, they could be a reservoir of ARGs. Surveillance of ARGs in the mangrove ecosystem has been largely carried out through a culturable approach. Bacterial isolates from shrimp farms in mangroves of Donghai Island (China) were resistant to multiple antibiotics such as tetracycline, trimethoprim, cefazolin, chloramphenicol, and ampicillin (Zhang et al., 2011). Similarly, MDR bacterial isolates under diverse phylogenetic groups isolated from Malaysian and Vietnamese shrimp farms were also resistant to β -lactam, aminoglycoside, sulphonamide, and tetracycline antibiotics (Jalal et al., 2010; Le et al., 2005). MDR isolates have also been

identified in Indian mangroves (Kerala and Sundarban) dominated by the Bacillaceae family having strong biofilm formation properties (Bhattacharyya et al., 2019; Imchen et al., 2019). Microarray based studies using GeoChip assay have also detected a similar predominance of MDR genes in the National Mangrove Nature Reserves of South China (Liu et al., 2020). Such an abundance of MDR isolates in mangroves could be due to aquaculture and other anthropogenic activities since commonly used antibiotics in aquaculture were detected in Gaoqiao mangrove (China) sediments and a strong association of ARGs towards waste input from different habitats (Li et al., 2016). Furthermore, ARGs in mangrove sediments and water had a significant positive correlation indicating that the ARGs could be mobile and disseminated through the water current (Zhang et al., 2011). For instance, *Escherichia coli* isolated from Matang mangrove (Malaysia), exposed to aquaculture and other anthropogenic activities, were also found to be MDR with the predominance of integron I and integron II (Ghaderpour et al., 2015). Likewise, the prevalence of the bla_{TEM} gene in Sundarban mangrove (India) had a positive correlation with anthropogenic influences (Bhattacharyya et al., 2019).

The prevalence of ARGs has been associated with misuse and overconsumption of antibiotics (Wang et al., 2017). Antibiotic usage per capita is also positively associated with the density of doctors and such services (Masiero et al., 2010). On the contrary, recent big data analysis has found the socioeconomic condition to have a stronger association rather than the antibiotic consumption (Collignon et al., 2018). An inverse correlation between antibiotic consumption and ARGs was observed in a recent study from wastewater treatment plants (Collignon et al., 2018). Unfortunately, the implementation of the socio-economic status of a country has been largely neglected as a factor for ARG prevalence (Collignon and Beggs, 2019). Furthermore, resistome studies on environmental sediments, particularly in mangroves, have received considerably less attention than the clinical settings. With the advent of next generation sequencing, resistome studies through WMS are preferred over traditional techniques such as PCR and microarray to overcome limitations on low throughput and amplification biases (J. Zeng et al., 2019). Recent metagenomic studies of ARGs in Chinese and Brazilian mangrove sediments have also detected multidrug resistance genes and mobile genetic elements (MGEs) (Cabral et al., 2016; Zhao et al., 2019). The WMS coupled with deep machine learning (ML) has pushed considerable advances in various aspects of biological research (Fiannaca et al., 2018). However, mangrove WMS data have never been analyzed with the ML approaches so far. In addition, despite the prevalence

of ARGs in various ecosystems, WMS resistome analysis using machine learning approach by the implementation of multiple databases is also lacking for data analysis. Hence, in the current study, we have implemented the DeepARG machine learning model based on Arango et al., (Arango-Argoty et al., 2018), to catalog a comprehensive list of ARGs in global mangrove sediments with the following hypothesis: (i) Diversity of ARG would be influenced not only by human intervention but also by geography (ii) Diversity of ARG could have a significant association to the socioeconomic status of the host country (iii) Based on the sedimented nature, mangrove sediment resistome would share higher similarities to terrestrial rather than ocean resistome despite their similarities in salinity.

MATERIALS AND METHODS

Data collection

Mangrove sediment metagenomes from Brazil, Saudi Arabia, China, India, and Malaysia were retrieved from NCBI SRA and MG-RAST which were sequenced using the whole metagenome sequencing (WMS) approach (Alzubaidy et al., 2016; Andreote et al., 2012; Imchen et al., 2018, 2017; Keegan et al., 2016; Li et al., 2019; Thompson et al., 2013; Zhao et al., 2019). The WMS data were based on next generation sequencing platform Illumina and Roche 454 (Luo et al., 2012). The metagenome data was further confirmed with their corresponding publications. The data on pristine and non-pristine nature as well as the sampling locations were obtained from the published articles or personal email communication(s) to the corresponding authors. For comparative study against other ecosystems, the terrestrial ecosystem was represented by WMS data originating from agriculture, urban area, forest, and grasslands (DeAngelis et al., 2010; Fierer et al., 2012; Kerfahi et al., 2019; Mendes and Tsai, 2018; Pouyat et al., 2017). The ocean ecosystem was represented by the WMS data from the Global Ocean Sampling expedition (GOS) metagenomes. The GOS expedition is one of the largest efforts to sequence the ocean microbiome with over 22 million genes identified from the project (Rusch et al., 2007). We implemented the MG-RAST pipeline for quality control and DeepARG v2.0 machine learning models for ARG prediction (Arango-Argoty et al., 2018). The DeepARG database is an amalgamation of unique ARGs from the Comprehensive Antibiotic Resistance Database (CARD), the Antibiotic Resistance Genes Database (ARDB), and Universal Protein Resource (UniProt) (Apweiler et al., 2004; Jia et al., 2016; Liu and Pop, 2009). The database, having a

total of 14933 unique ARGs under 30 ARG categories, was used for supervised deep learning model. The ARG sequences were represented with dissimilarity based classification (Sørensen et al., 2010) and trained by using ARGs sequences from UNIPORT against the CARD and ARDB ARG sequences. The model consists of 4333 units in the input layers, 4 hidden layers consisting of 3600 units, and 30 units in the output layers. Docker image of the DeepARG was pulled and the ARG prediction was set with at least 80% probability and minimum alignment coverage with an e value of less than 1×10^{-10} .

Data normalization, visualization, and analysis

The predicted ARG datasets were imported to phyloseq v1.3 in R (McMurdie and Holmes, 2013; Team, 2017). Alpha diversities (Observed, Chao1, Shannon, and Pielou's evenness) were calculated, from the rarefied data, with the *alpha* function from the microbiome v1.8 package (Lahti et al., 2017; Weiss et al., 2017). For comparative analysis, raw datasets were initially batch corrected with preprocessCore v1.48 as described in the literature (Bolstad and Bolstad, 2015; Medina et al., 2019). The distribution of ARG classes and subtypes were visualized using *ggplot* and *plot_bar* functions from ggplot2 v3.3 and phyloseq (Wickham, 2016). Venn diagram was prepared using the online tool InteractiVenn and InkScape v0.92 (Heberle et al., 2015; Schöler and Schöler, 2014). Weighted and unweighted UniFrac PCoA was plotted with the *plot_ordination* function under the phyloseq package. ARG class and subtype heatmaps were generated through the *plot_heatmap* function. The data on Human development index (HDI), antibiotic consumption (AC), GDP per capita, and Population were retrieved from The Global Bank Group (<http://hdr.undp.org/en/data>) and ResistanceMap (The Center for Disease, Dynamics Economics & Policy, 2020). HDI is implemented by the United Nations to measure a country's development. It factors three important indexes of development: life expectancy (I_{Health}), access to education ($I_{\text{Education}}$), and per capita income (I_{Income}) (United Nations Development Programme, 2019). It is calculated by aggregation of the three indices as follows (Roser, 2014): $\text{HDI} = (I_{\text{Health}} * I_{\text{Education}} * I_{\text{Income}})^{1/3}$

Statistical analysis

Analysis of similarities (ANOSIM) was calculated with 999 permutations in R vegan package (Oksanen et al., 2007). Differentially abundant ARG subtypes (>1% relative abundance) were screened using the Kruskal Wallis test, *p* values of less than 0.05 were considered statistically significant. Correlation between ARG diversities and socio-economic status of the host countries was determined with linear *r* (Pearson) correlation and corrected with Bonferroni

correction in PAST v3.25 (Hammer et al., 2001). For all statistical tests, a p -value of less than 0.05 was considered significant.

RESULTS

ARG diversity

Chao1 estimation of the ARGs ranged from 72.34 in CS02 (Saudi Arabia) to 183.72 in MAL (India) (Table S1). Human intervened mangrove sediments had significantly higher alpha diversities (Observed, Chao1, Shannon index, and Pielou's evenness) than the pristine mangrove sediments ($p<0.05$) (Figure 1A-D). The highest ARG diversity was observed in anthropogenic affected mangrove in Baia (Brazil) followed by the datasets from Indian mangroves with moderate anthropogenic activities. Similarly, the pristine mangrove dataset, SYP and HKP, had lower Observed and Chao1 diversity than the non-pristine counterpart SYC and HKC. The differences in ARG diversity between the countries were also statistically significant ($p=0.02$) (Figure 1E, F). India had the highest diversity of ARGs while Saudi Arabia had the lowest ARG diversity. Similarly, the Shannon diversity index (H) (Figure 1G) and Pielou's evenness (J) (Figure 1H) were also highest ($H=3.74$; $J=0.87$) in PGD (India) and lowest ($H=2.86$; $J=0.72$) in CS02 (Saudi). ARG distribution based on geography ($R_{ANOSIM}=39.26\%$; $p<0.005$) was found to have a stronger influence than the pristine nature ($R_{ANOSIM}=16.92\%$; $p<0.005$). Venn diagram representation showed that all ARGs were shared between at least two countries. However, only 38.76% of the resistome were common to all the countries (Figure 2A). The difference in ARG distribution between the countries was further confirmed in the Unweighted and Weighted UniFrac PCoA plot (Figure 2B, C). Correlation analysis between ARG diversity against Human development index (HDI), antibiotic consumption (AC), GDP per capita, and Population retrieved from The Global Bank Group (<https://data.worldbank.org/>) indicated a significant association of HDI and antibiotic consumption ($R=0.56$; $p=0.02$) (Table 1). However, HDI had an inverse correlation to Observed and Chao1 ARG diversities ($R=-0.7$ and $R=-0.53$; $p<0.001$ respectively). HDI was also negatively correlated to the Shannon index ($R=-0.8$; $p<0.001$) and Pielou's evenness ($R=-0.79$; $p<0.001$). We also observed that a higher population in the host country was associated with lower antibiotic consumption ($R=-0.74$; $p<0.001$) and HDI ($R=-0.68$; $p<0.001$) but high Shannon diversity ($R=0.58$; $p=0.01$).

Antibiotic resistant class distribution

The mangrove resistome consisted of 27 ARG classes (Figure 3). Pristine as well as human intervened mangrove datasets from China (M_Ch_SRX4104976 and M_Ch_SRX4104977), India (M_In_MAL), and Malaysia (M_Ma_VirginForest and M_Ma_Penang) harbored all the 27 ARG classes. The least diversity of antibiotic resistance class was observed in datasets derived from pristine mangroves i.e. M_Br_Mgv-4 (n=4), M_Sa_CS02 (n=14), and M_Sa_CS01 (n=15). This indicates that ARG diversity varies substantially irrespective of pristine nature. The mangrove resistome was consistently dominated by multidrug resistance (52.98±0.87%), and, “Macrolides, lincosamides, streptogramins” (MLS) (13.46±0.38%) (Table S2). The majority of the multidrug resistance was represented by multidrug efflux pumps, which were significantly enriched in human intervened mangroves (Figure S1-S3). In addition, mobile ARGs such as *vanA* (vancomycin resistance), *tetA(48)* (tetracycline resistance), *mcr1-5* (mobilized colistin resistance), and *tetX* were also detected (Table S3; Figure S4-S6). The resistome was composed of 709 ARG subtypes (Table S3). Three resistant subtypes viz. *rpoB2* (19.26±0.01%) (rifampin-resistant beta-subunit RNA polymerase), multidrug ABC (ATP-binding cassette) transporter (10.40±0.23%), and *macB* (8.84±0.36%) (macrolide resistance) were predominant, with the *rpoB2* being the most dominant in all datasets. This was followed by *tetA* (4.13±0.35%), *mexF* (multidrug efflux) (3.26±0.19%), *CpxR* (conjugative plasmid gene expression; RND - resistance nodulation division - antibiotic efflux pump) (2.93±0.20%), *bcrA* (bacitracin transport ATP-binding protein) (2.38±0.24%), *acrB* (Acridine resistance protein; multidrug efflux pump subunit) (2.37±0.18%), *mexW* (2.19±0.17%) and *vanR* (1.99±0.11%), which jointly dominated more than half (57.75±2.01%) of the resistome. There were also several dominant subtypes (>1% relative abundance) significantly different among the countries (Figure S7). Datasets from Indian mangroves had the highest number of significantly enriched ARGs such as *cmeB* (*Campylobacter* multidrug efflux), *emrA-B* (*Escherichia coli* multiple antibiotic resistance regulator), *kdpE* (potassium-dependent protein E; resistance to aminoglycoside antibiotic), *mdtB* (transporter gene in multidrug transporter ABC), *ugd* (*PmrE*) (resistance to peptide antibiotic), and *TriC* (triclosan resistance). Meanwhile, Brazil had the highest abundance of *efrB* (*Enterococcus faecalis* multidrug resistance) and *PmrF* (resistance to peptide antibiotic), while *rosB* (regulation of O₁₅₇ antigen synthesis; resistance to peptide antibiotic) and *smeE* (*Stenotrophomonas* multiple efflux) in China. Datasets from Saudi Arabia were significantly enriched with *tetA*; however, it had a significantly lower abundance of *macB*.

Comparative analysis against other ecosystems

Mangrove datasets were further compared against terrestrial and ocean ecosystems to investigate their similarities and differences in resistome. The Chao1 and Shannon Index varied significantly between the three ecosystems. The ocean ecosystem harbored the lowest ARG diversity ($S_{\text{chao1}}=71.12$) followed by mangroves ($S_{\text{chao1}}=258.07$) and terrestrial ($S_{\text{chao1}}=294.07$) ecosystems (Figure 4A, B). However, the mangrove ecosystem had the lowest evenness (0.7 ± 0.03) followed by terrestrial (0.74 ± 0.03) and ocean (0.86 ± 0.01) indicating that, compared to terrestrial and ocean ecosystems, the mangrove resistome could be dominated by certain ARGs to a greater extent (Figure 4C). Heatmap analysis of the resistant classes ($n=28$) showed that multidrug resistance dominated all three ecosystems (Figure 4D). Resistance against glycopeptide, aminoglycosides, diaminopyrimide, phenicols, and sulfonamide was markedly over-represented in ocean datasets (Table S4). The terrestrial datasets were over-represented with nucleosides, triclosan, fosmidomycin, and rifamycin. On the other hand, mangrove ecosystems were over-represented by MLS and tetracycline resistance classes, which were represented slightly lower in the terrestrial ecosystem. Further insights into the subtypes showed a similar pattern between mangrove and terrestrial sediments compared to the ocean (Figure 5). There were also several ARGs that were shared between the ocean and terrestrial. Nonetheless, the dominant ARGs in mangroves were in common to that of terrestrial resistome, which is also in agreement with Weighted UniFrac PCoA observations (Figure 6A). The ARG subtypes in ocean datasets varied remarkably to that of mangrove and terrestrial. ARG subtypes such as *abeS* (*Acinetobacter baumannii* efflux pump of SMR - small multidrug resistance - family) and *qacG* (quaternary ammonium compounds) were predominant in ocean datasets (Figure 6B). The other dominant subtypes in the ocean such as *ksgA* (kasugamycin resistance) and *ompR* (outer membrane protein R; multidrug resistance) were detected several folds (13.9 and 2.9 respectively) lower in mangrove and terrestrial. Likewise, several ARGs detected in mangrove and terrestrial datasets such as *rpoB2*, ABC transporter, and *macB* were undetected in the ocean datasets.

DISCUSSION

Influence of anthropogenic activities and ARG diversity in mangrove sediments

Antibiotic resistance genes are known to be of ancient origin that originated in the environmental microbes and disseminated to diverse phylogenetic groups before the clinical

usage of antibiotics (Wang et al., 2017). The prevalence of ARGs have been accelerated by the usage of antibiotics as additives in animals and aquaculture feeds to promote growth and ward off infections in the livestock for several decades (Yin et al., 2015). In addition to the undesirable consequence of excess antibiotic usage in agriculture and animal farming as well as in healthcare, anthropogenic activities such as waste disposal also enrich ARGs (Yang et al., 2018). It is, therefore, no wonder that the diversity of ARGs in non-pristine (human intervened) mangrove sediments were significantly higher ($p<0.05$) compared to pristine mangrove sediments (Figure 1A, B). Furthermore, the lowest ARG diversity was observed among the pristine samples (Table S1). The detection of ARGs in the pristine mangrove sediments could be attributed to the multiple roles of antibiotics such as in transcription modulation, SOS response, and interspecies competition (Sengupta et al., 2013). In addition, diversified microbiota in mangrove sediments leads to synergetic as well as antagonist microbial interactions in quest of food and space. Such interaction leads to the secretion of inhibitory bioactive molecules that suppresses competitors. Most of the clinically relevant antibiotics that are currently in the market are of such natural origin (Clardy et al., 2006). Resistance mechanisms to the inhibitory effects of such biomolecules are encoded in the producer organism to protect themselves. The competing microbes could also undergo a spontaneous mutation that leads to resistance against such inhibitory biomolecules (McLeod et al., 2018). Similarly, multidrug resistance has been identified from pristine soil in the Arctic and Antarctica (Kudinova et al., 2019). In line with this, within this study, some of the datasets from pristine (SRX4104977, SRX4104976, and VirginJungleForest-3) as well as non-pristine (MAL, BatuMaung_Penang) mangroves harbored equally high diversity of antibiotic resistant class ($n=27$) (Table S2). In addition to the increased Chao1 and Observed alpha diversities on non-pristine mangrove sediments, there was also a significant increase in Shannon diversity and Pielou's evenness (Figure 1C, D). Such an increase in alpha diversities shows that that human intervention enriches a large set of ARGs. The enrichments of ARGs in human intervened mangrove could lead to accelerated dissemination of ARGs between microbes of different phylogeny through horizontal gene transfer (Laroche et al., 2009). The degree of ARGs dissemination is quite rapid. This is evident from the detection of bla_{NDM-1} in a remote location of High Arctic soil within 3 years from the initial detection of bla_{NDM-1} in India (McCann et al., 2019).

Influence of host country in mangrove sediments ARG diversity

Differences in ARG diversities varied significantly based on human intervention as well as the host country ($p < 0.05$) (Figure 1A-H, 2A-C). ANOSIM analysis indicated that the host country had a higher influence ($R_{\text{ANOSIM}} = 39.26\%$; $p < 0.005$) rather than human intervention ($R_{\text{ANOSIM}} = 16.92\%$; $p < 0.005$). The differences in ARG diversity between the countries could be due to variations in several aspects such as antibiotic consumption, HDI, regulations, climate, etc. HDI has been observed to be positively associated with higher antibiotics consumption and a strong predictor variable for higher antibiotic consumption in Sao Paulo, Brazil (Kliemann et al., 2016). Similarly, in this study, we observed a strong positive correlation ($R = 0.56$; $p = 0.02$) between HDI and antibiotic consumption (Table 1). The positive association of HDI and antibiotic consumption could be due to several underlying factors. Antibiotic consumption is associated with a higher level of education, higher GDP, and the density of healthcare clinics in the area (Collignon et al., 2018; Klein et al., 2018; Kliemann et al., 2016). Furthermore, a recent analysis of global internet usage identified that the relative search volume (RSV) of probiotics was associated with antibiotic consumption (Mikołaj et al., 2019). Such observations indicate that antibiotic consumption increases with increased accessibility and affordability to antibiotics. Since, HDI is an index that accounts for GDP, education, and life expectancy (health), such an association could explain the association of HDI and antibiotic consumption. The positive association of HDI with increased antibiotic consumption is alarming because several studies have shown a direct relationship between antibiotic consumption and the escalation of antibiotic resistant isolates (Wojkowska-Mach et al., 2018). For instance, a positive correlation between antimicrobial usage and development of resistance to extended-spectrum cephalosporins and ciprofloxacin-resistant *Klebsiella pneumoniae* was observed in the university hospital in Taiwan (Hsueh et al., 2005). Similarly, a positive association of resistance to imipenem (carbapenems) and levofloxacin (quinolones) was also observed in *Acinetobacter baumannii* and *Enterococcus faecium* respectively (S. Zeng et al., 2019). Despite the empirical evidence of a positive association between ARG diversity to antibiotic consumption, a negative correlation has also been observed between cefotaxime-resistant *E. coli* and the application of cefotaxime (Lai et al., 2011). Moreover, recent studies have reported a stronger influence of the socioeconomic status of the host country rather than the antibiotic consumption (Collignon et al., 2018; Deschepper et al., 2008; Hendriksen et al., 2019). Although the prevalence of certain ARG class was associated to antibiotic usage of the particular class, the overall ARG diversity has

no association to overall antimicrobial usage (Hendriksen et al., 2019) In fact, antibiotic consumption had a significant inverse correlation to ARGs prevalence (Collignon et al., 2018). In line to this, we have also observed a negative correlation between ARG diversity and antibiotic consumption ($R=-0.613$; $p<0.05$) for the first time in global mangrove sediments (Table 1). Furthermore, ARG diversity was also inversely correlational to HDI ($R=-0.805$; $p<0.05$). This indicates that, besides the enrichment effect of ARGs due to antibiotic over-usage, socioeconomic factors could have a strong effect on the ARG dissemination. It is hypothesized that contagion, rather than antibiotic consumption, could be a major factor for ARGs prevalence which is dependent on various social and economic attributes (Collignon and Beggs, 2019). This is further supported by the increased incidence of drug-resistance in third world countries with lower gross income (Malik and Bhattacharyya, 2019)

The prevalence of ARGs in mangrove sediments is clinically relevant since the incidence of ARGs in clinical isolates has been reported to be associated with ARGs in aquaculture (Reverter et al., 2020). Aquaculture in the mangrove environment is a common practice worldwide. Although overuse of antibiotics is known to selectively enrich ARGs, it is magnified by poor sanitation, poor regulation, malnutrition, etc. which is common in countries with low HDI or low-to-middle income countries (LMIC). Regulation on antibiotic usage in agriculture and animal farms are also weaker in LMIC (Mensah et al., 2014). LMIC with budgetary constraints lacks preparedness, adequate water, sanitation, ARG surveillance, infrastructure, food safety, and proper treatment of waste (Iskandar et al., 2020). For instance, waste from slaughterhouses is released directly into the environment without pre-treatment (Iskandar et al., 2020). Antibiotic treatments in outpatient are also often abandoned when the symptoms subside before the full course which often leads to the emergence of resistant strains (Ayukekbong et al., 2017; Okeke et al., 1999). Traditional treatments with herbal combinations are often composed of unknown substances and potency which could contribute to resistance development (Ayukekbong et al., 2017). Antimicrobials in personal hygiene can also serve as ARG selective agents (Meek et al., 2015). LMICs are also known to have poor quality medicine due to counterfeit, expiry, or lack of proper storage that leads to suboptimal dosage (Ayukekbong et al., 2017). Lack of proper vaccination could also lead to infections that drive higher usage of antimicrobials (Diaz et al., 2018). Rural areas in LMIC are also prone to infections and the dissemination of ARGs between environments. For instance, livestock such as poultry is reared near to home which increases the risk of ARGs

from livestock to humans (Iskandar et al., 2020). Untreated waste from livestock is also used for various purposes such as fertilizers (Iskandar et al., 2020). There are also instances of sharing a common water source for bathing and cleaning where animals are left free for grazing, drinking, and defecating (Finley et al., 2013).

Previous studies have also indicated the complex relationship between antibiotic use and antibiotic resistance. The selective enrichment of ARGs could also vary between the types of antibiotics, types of fitness cost, and fitness cost variation between strains (Friedman and Whitney, 2008). To overcome or to reduce the emergence of ARGs, several steps should be taken up with utmost importance. In the clinical settings, unnecessary prescription of broad-spectrum antibiotics by physicians should be avoided (Lee et al., 2013). The hospital environment should also be disinfected regularly and healthcare workers should also be provided with adequate supplies to maintain personal hygiene. Strict governance also ensures better functioning of healthcare systems and ARG surveillance (Pokharel et al., 2019). Plasmid curing or inhibition of plasmid conjugation in bacterial populations such as in the gut microbiome of farm animals and wastewater treatment plants could reduce the dissemination of ARGs (Buckner et al., 2018). Recent advancements in the CRISPR-Cas system is also anticipated to provide strain-specific selectivity to combat the ARG crisis (Gholizadeh et al., 2020). The spread of ARGs could also be based on social patterns that include lifestyle and cultural aspects. Hence, multidisciplinary research should be encouraged such as social research and mathematical modeling (Knight et al., 2018). Furthermore, in developing countries, self-medication is quite common when formal prescription from a physician is not affordable which could lead to the development and dissemination of resistant strains (Morgan et al., 2011). Hence, affordable healthcare to the general public in rural areas should be set up and educate the public as well as healthcare workers on the wise use of drugs.

Distribution of ARGs subtype

Mangrove resistome was represented by 27 antibiotic resistant classes and 709 ARG subtypes. The resistome was dominated by 10 major subtypes that covered $57.75 \pm 2.01\%$ of the resistome (Table S3; Figure 3). The dominant subtypes were of three mechanisms viz. mutations in the antibiotic target (*rpoB2*), two-component systems, and efflux pumps (RND, ABC, and MFP - major facilitator superfamily). *rpoB2* is a mutant copy of the RNA polymerase β -subunit gene which confers resistance to rifamycin class antibiotics (Hoshino et al., 2010). In addition, antibiotic resistance through two component systems under the OmpR

family *CpxR* ($2.88 \pm 0.24\%$) and *vanR* ($1.99 \pm 0.11\%$) were also over-represented in the metagenome. The *CpxR* in conjunction with *CpxA* regulates several functions, in addition to drug resistance, such as stress response to alkaline pH, misfolded proteins, periplasmic stress, etc. (Weatherspoon-Griffin et al., 2014). Although mutations in antibiotic target and two component systems represented a substantial portion of the resistome ($>20\%$), multidrug efflux pumps under RND, MFP, and ABC covered over one-third of the resistome (Figure S1, S2). Furthermore, the multidrug efflux pumps were significantly ($p < 0.05$) enriched in the human intervened mangroves (Figure S3). The enrichment of multidrug efflux pumps could have multiple roles in addition to drug resistance because they also actively perform an influx of essential nutrients and efflux of toxic compounds (Ginn et al., 2000). They are also present in all the domains of life (Locher, 2016). Owing to the broad specificity of the efflux pumps, it confers cross-resistance to multiple antibiotics. Efflux pumps are known to confer a certain level of resistance to all types of antibiotics (Morar and Wright, 2010). Efflux of antibiotics also reduces the intracellular concentration to an extent that prolongs the survival time which in turn creates a window for the possible development of antibiotics resistance through spontaneous mutation (Piddock, 2006). Hence, irrespective of their roles, they represent a major hurdle in the treatment of bacterial infections and have also been proposed as a suitable target for therapeutic applications (Kumar et al., 2013). There were also several ARG subtypes enriched in pristine mangrove sediments. Determinants that confer resistance to glycopeptide and tetracycline were significantly higher in the pristine mangroves. Coincidentally, resistance to glycopeptide and tetracycline were also reported in pristine ancient permafrost (Dcosta et al., 2011). Tetracycline antibiotics are polyketides derived from natural secondary metabolism in Actinomycete. Several microbes are also capable of utilizing antibiotics as sole carbon source (Dantas et al., 2008). It is estimated that the resistance to tetracycline and glycopeptide could have existed for millions of years (Balassiano et al., 2007; Baltz, 2007). There was a high prevalence of glycopeptide ($n=33$) and tetracycline resistant subtypes ($n=47$) with *tetA*, *tetA(48)*, *vanR*, and *vanS* being the most dominant in pristine as well as human intervened metagenomes (Figure S4, S5). *tetA*, an efflux pump, is one of the most common resistant genes in soil, food, and particularly in the isolates from clinical samples (Park et al., 2010). On the contrary, the resistance mechanism based on tetracycline inactivation has been rarely detected in environmental metagenomes (Forsberg et al., 2015). In this study, the detection of *tetA(48)*, a flavoenzyme protein that can deactivate tetracycline antibiotics, as one of the dominant tetracycline resistant determinants is worrisome. Furthermore, *vanA* and *tetA(48)* are known to be mobile (Faron et al., 2016;

Forsberg et al., 2015). We also identified *mcr1-5* and *tetX* dominated in the human intervened mangrove metagenome, which are known to be mobile and confers resistance to last resort antibiotic colistin and tigecycline (Figure S6) (Yang et al., 2004). The wide prevalence of chromosomal and mobile ARGs in the metagenome taken together with higher microbial diversity would lead to horizontal gene transfer to pathogens that could cross the last line of defense.

Higher ARGs diversity in sediment samples

To understand the uniqueness of mangrove resistome, we compared it to terrestrial and ocean resistomes. The ocean resistome had substantially lower diversity ($S_{\text{chao1}}=71.12$) than the mangroves ($S_{\text{chao1}}=258.07$) and terrestrial ($S_{\text{chao1}}=294.07$) resistome (Figure 4A). Shannon index followed a similar trend (Figure 4B). Although ocean and mangrove share a similar marine (saline) environment, the higher similarity between mangrove and terrestrial was observed among the ARG class and subtypes (Figure 4D,5) indicating that it is the sedimented nature that determines the ARG diversity. Soil sediments are a known reservoir of ARGs acquired from anthropogenic activities as well as naturally occurring resistant genes (McCann et al., 2019). The higher diversity of ARGs in mangrove sediments and terrestrial could also be explained according to the particle attached nature of the microbes. It is established that particle attached bacteria have a stronger arsenal of defense mechanisms, which are required for colonization of the particle and defend from competitors. Such colonization of particles also contributes to the essential ecological process of degradation and biogeochemical cycles (Long and Azam, 2001). However, the lower abundance of ARGs in mangrove compared to terrestrial could be due to the high antagonistic interactions in the mangrove microbial community (Imchen et al., 2019). In addition, previous reports have also indicated a lower abundance of ARGs in mangrove wetlands (Zhao et al., 2019). Although the diversity of ARGs in the ocean was lower than the other contrasting ecosystems, it should be noted that several other water bodies such as wastewater treatment plants have a significantly higher ARG abundance (Barancheshme and Munir, 2018). The most abundant resistant determinant in Ocean, *ksgA* ($10.19 \pm 0.74\%$) that encodes resistance to aminoglycoside antibiotic kasugamycin, was several folds higher than mangroves ($0.73 \pm 0.09\%$) and terrestrial ($0.12 \pm 0.02\%$) (Figure 6A). On the contrary, the most abundant resistant subtype (*rpoB2*) in terrestrial ($12.75 \pm 0.01\%$) and mangrove ($13.9 \pm 0.01\%$) was not detected in the ocean metagenome. Similarly, several dominant ARGs in mangroves were shared commonly to terrestrial datasets, which is also in agreement with Weighted UniFrac

PCoA observations (Figure 6B). The high diversity of ARGs in mangroves as well as other ecosystems could lead to an outbreak of multidrug-resistant bacterial infections.

CONCLUSIONS

ARGs are widespread in environmental settings which have turned into a global crisis in the medical, agricultural, and farming sectors. In this study, we have analyzed the resistome of global mangrove sediment through the whole metagenome approach with the aid of deep machine learning to investigate the influence of anthropogenic activities, its association with socioeconomic factors, and comparative analysis against contrasting ecosystems. Datasets from human intervened mangrove sediments have a significantly higher abundance of multidrug efflux pumps. Nonetheless, ARG subtypes under glycopeptide (n=33) and tetracycline resistant (n=47) were prevalent in pristine as well as human intervened mangrove sediments. Multidrug resistance attributed to the efflux pumps, mutations in target genes, and two component systems over-represented the resistome. Resistance to the last line of antibiotics, colistin and tigecycline, conferred by mobile antibiotic resistance determinants such as *tetX* and *mcr* were also detected. ARG diversity was inversely correlated to HDI as well as antibiotic consumption ($p<0.05$) indicating that the spread of ARGs could be majorly linked to weaker socioeconomic conditions rather than the consumption of antibiotics alone. Further comparison to terrestrial and ocean ecosystems shows the potential role of terrestrial and mangrove as ARG reservoirs. Our study shows that the mangrove ecosystems are adversely affected by the anthropological intervention and the HDI could be a potential predictor of resistome, not only in clinical settings but also for environmental settings.

DECLARATIONS

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490 mgp2996, mgp12156, mgp18555, mgp16346

491 **Authors contributions.** Conceived and designed the experiments: RK and IM. Analyzed the
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FIGURE LEGENDS

Figure 1: A significant higher [A] Observed, [B] Chao1, [C] Shannon diversity and [D] Pielou's evenness were observed in the human intervened metagenome. [E-H] Differences in alpha diversities were also observed among the countries.

Figure 2: [A] Common ARGs between the countries represented in Venn diagram showed a higher proportion of the ARGs being shared between at least two countries but only 38.76% among all the countries. [F] Unweighted and [G] weighted UniFrac PCoA plot showed moderate difference in diversity and abundance among the countries.

Figure 3: Stacked bar plot of antibiotics resistant class in the mangrove metagenome showed consistent similarities in dominance of multidrug, MLS and tetracycline at a global scale.

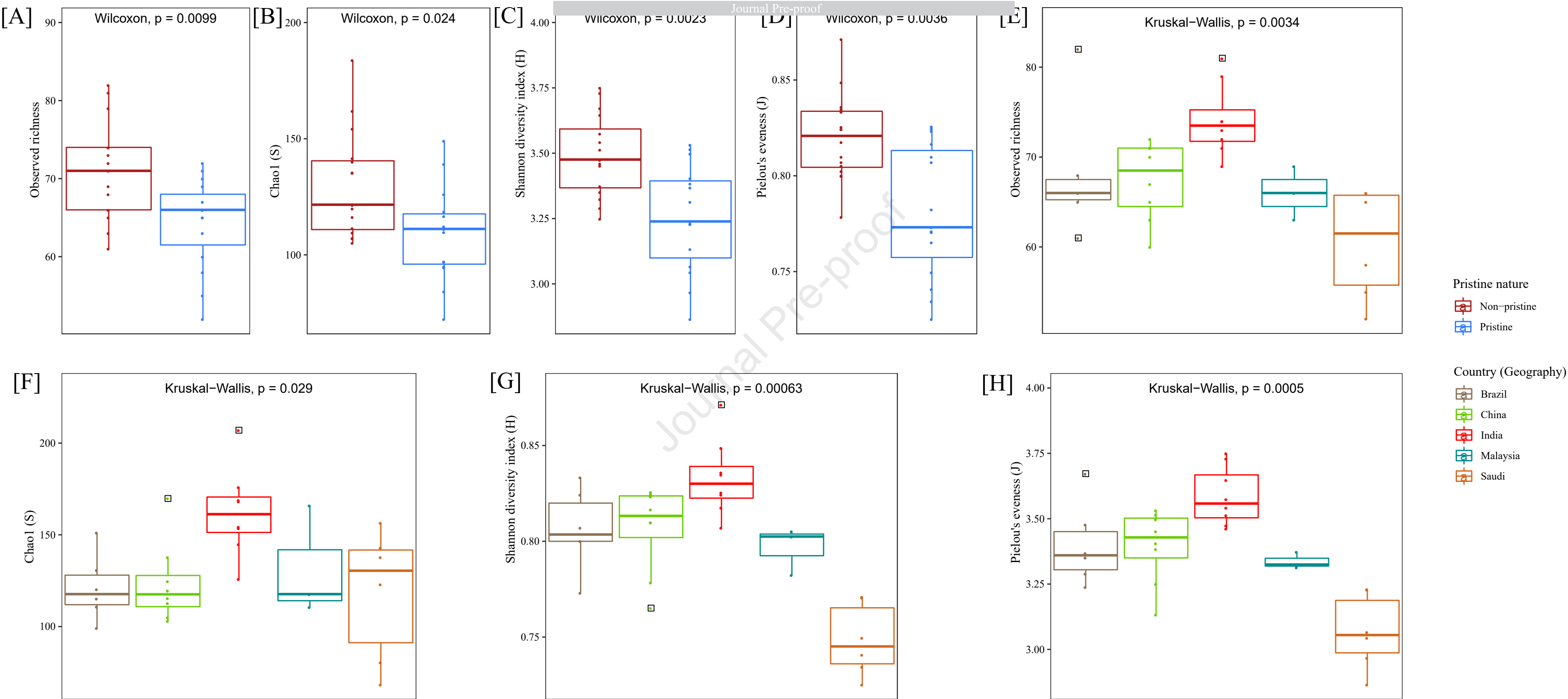
Figure 4: Comparative analysis between three contrasting ecosystems i.e. mangrove, ocean and terrestrial, showed ocean with the lowest [A] Chao1 and [B] Shannon diversity index. [C] However, mangrove exhibited the lowest Pielou's evenness. [D] Heatmap analysis of the ecosystems further shows higher enrichment of ARG class in terrestrial datasets.

Figure 5: Heatmap analysis of the antibiotics resistant subtypes confirms the higher diversity of ARGs in the terrestrial datasets while sharing similarities to mangrove datasets among the dominant ARGs.

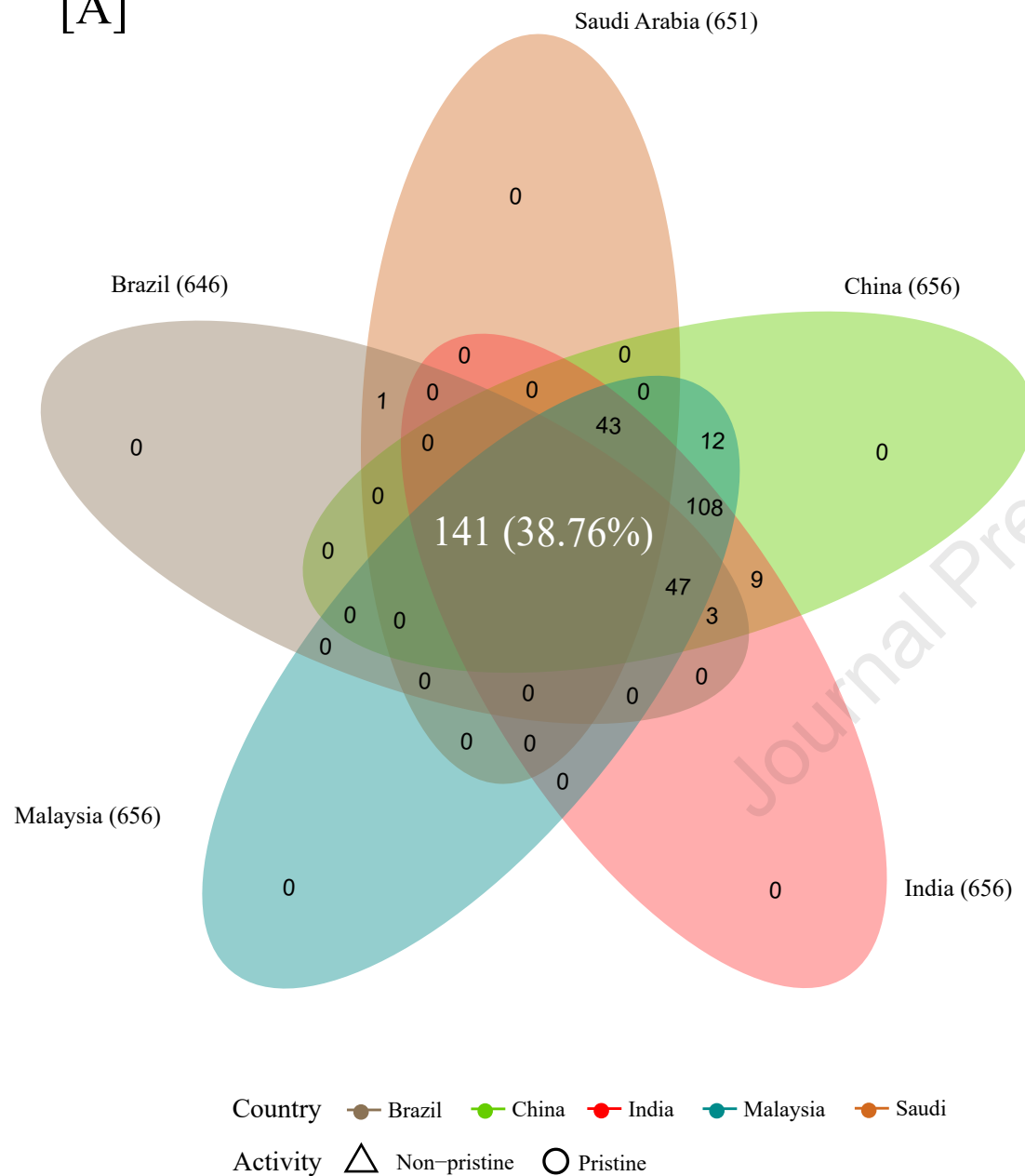
Figure 6: [A] Weighted UniFrac PCoA plot clustered ocean resistome distantly from the mangrove and terrestrial datasets. Furthermore, the extensive diversity of ARGs in the terrestrial is indicated by the wider spread of the samples compared to mangroves. [B] The dominant ARGs in the ecosystems visualized in stacked bar showed higher similarities between mangrove and terrestrial.

Table 1: Correlation between antibiotic resistance gene (ARG) diversity and host country's socioeconomic factors such as human development index (HDI), antibiotics consumption (AC) and population indicated that ARG diversity is inversely proportional to HDI. The factors that showed significant correlated are indicated in star (*) mark.

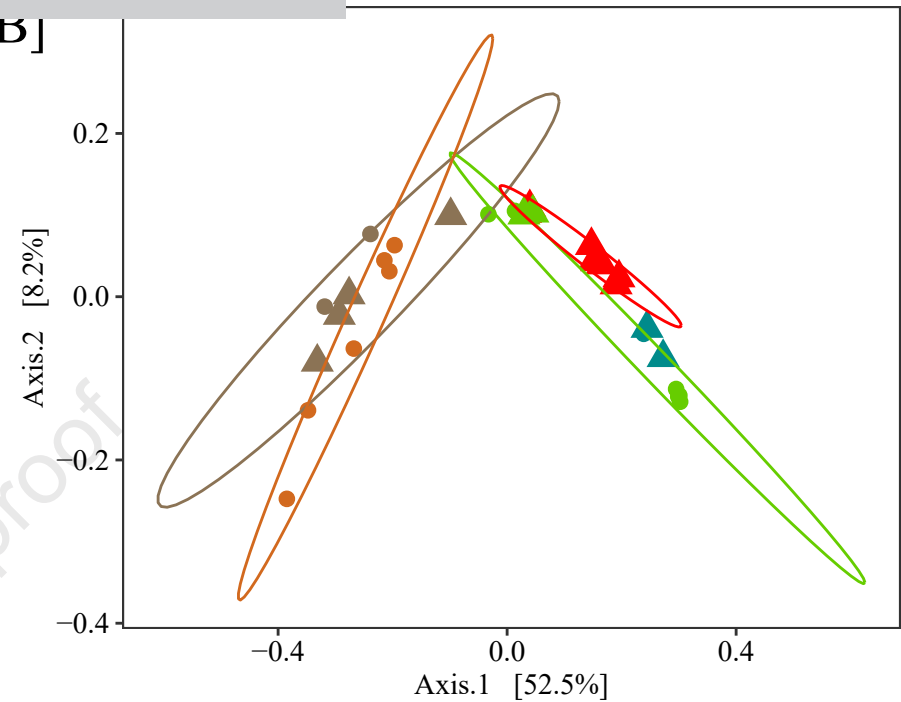
<i>Correlation between</i>	<i>R</i>	<i>p value</i>
<i>AC vs. Population*</i>	-0.745	3.2×10^{-4}
<i>AC vs. HDI*</i>	0.562	0.02
<i>AC vs. Chao1</i>	-0.116	1
<i>AC vs. Shannon diversity*</i>	-0.613	0.005
<i>AC vs. Pielou's evenness*</i>	-0.657	0.001
<i>AC vs. Observed richness</i>	-0.446	0.246
<i>Population vs. HDI*</i>	-0.688	3.9×10^{-4}
<i>Population vs. Chao1</i>	0.201	1
<i>Population vs. Shannon diversity*</i>	0.582	0.012
<i>Population vs. Pielou's evenness*</i>	0.593	0.9×10^{-3}
<i>Population vs. Observed richness*</i>	0.477	0.137
<i>HDI vs. Chao1*</i>	-0.533	0.041
<i>HDI vs. Shannon diversity*</i>	-0.805	0.9×10^{-8}
<i>HDI vs. Pielou's evenness*</i>	-0.799	1.4×10^{-6}
<i>HDI vs. Observed richness*</i>	-0.707	1.8×10^{-4}



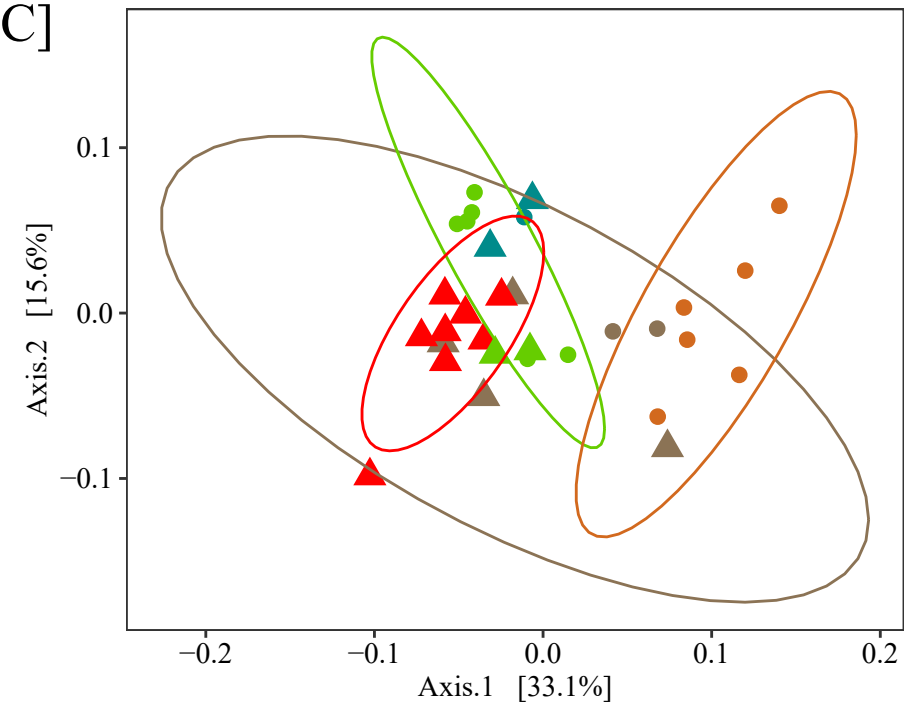
[A]

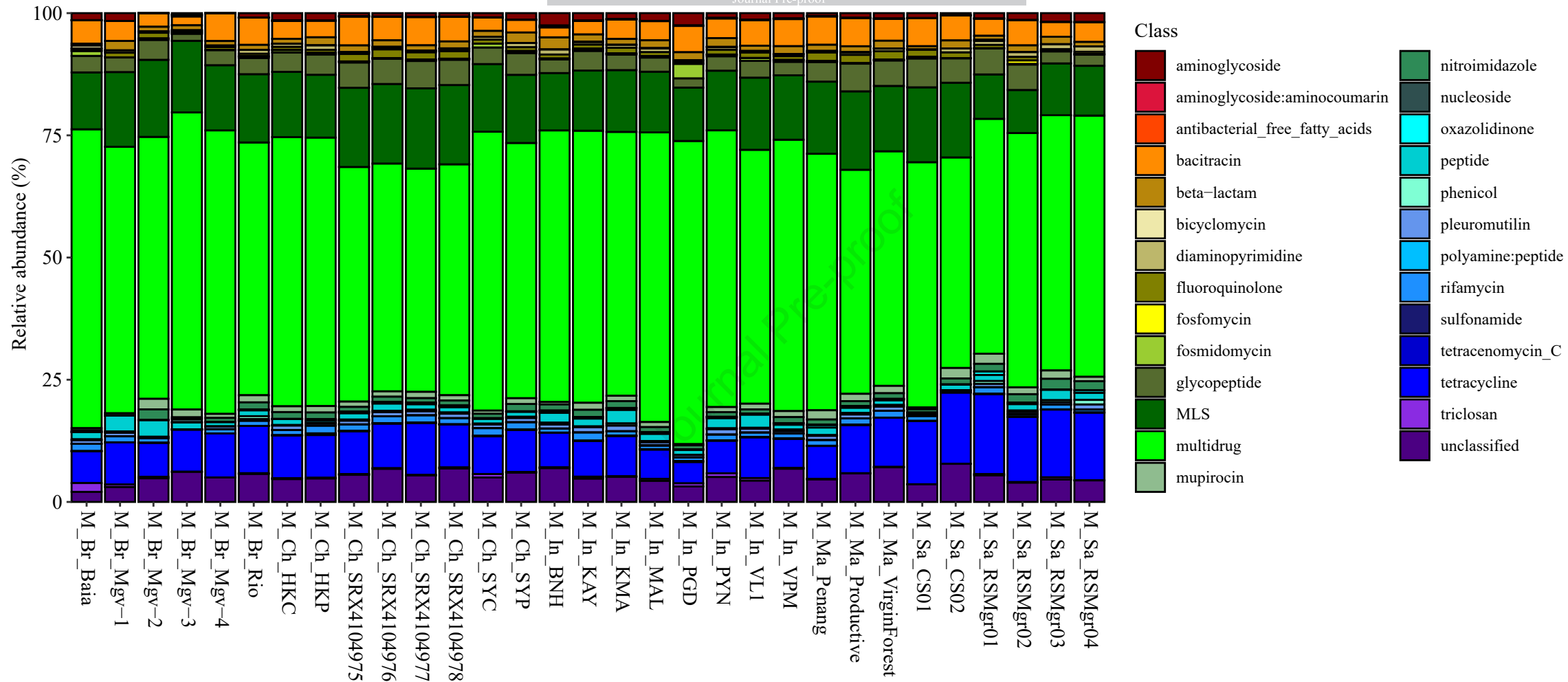


[B]

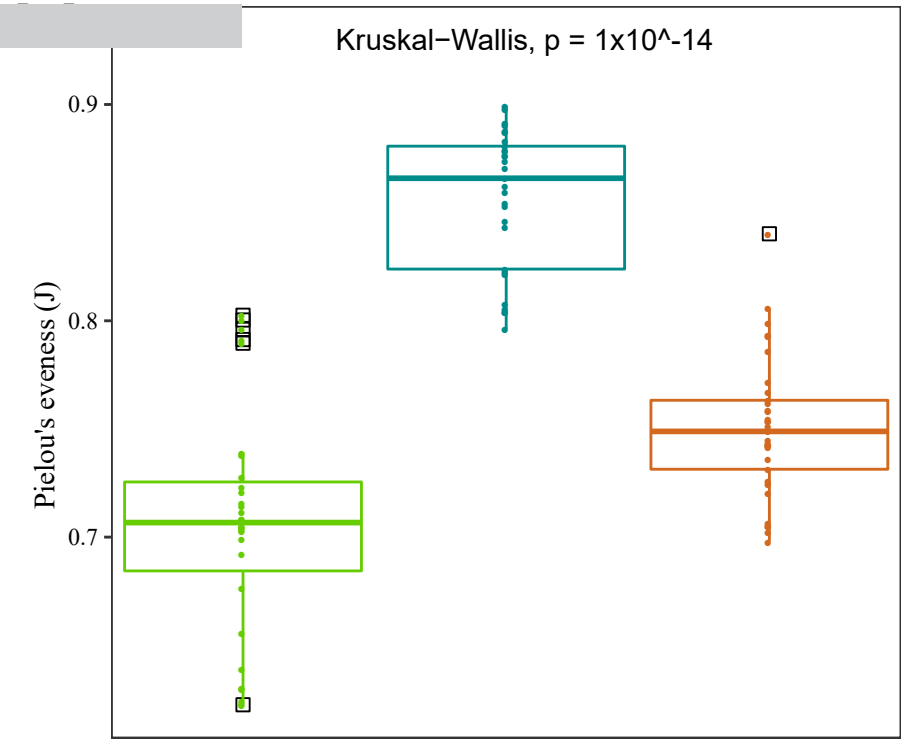
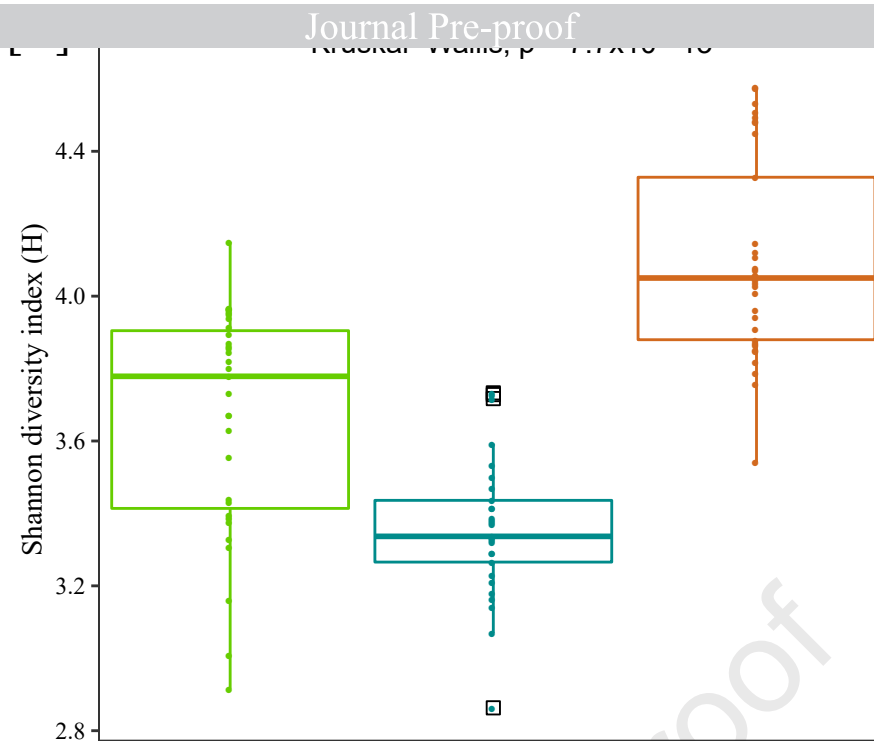
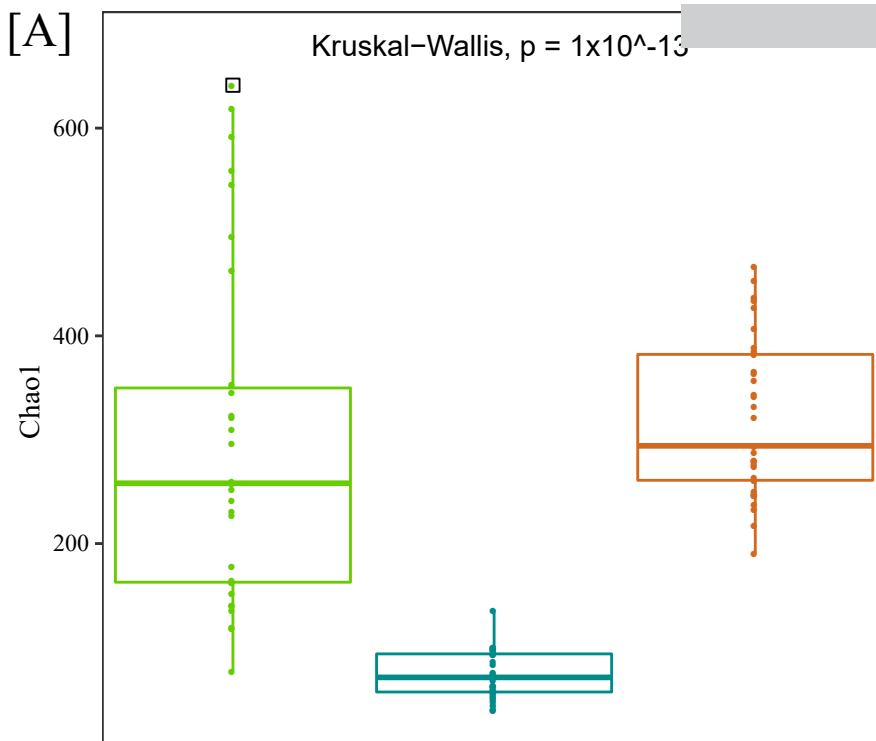


[C]

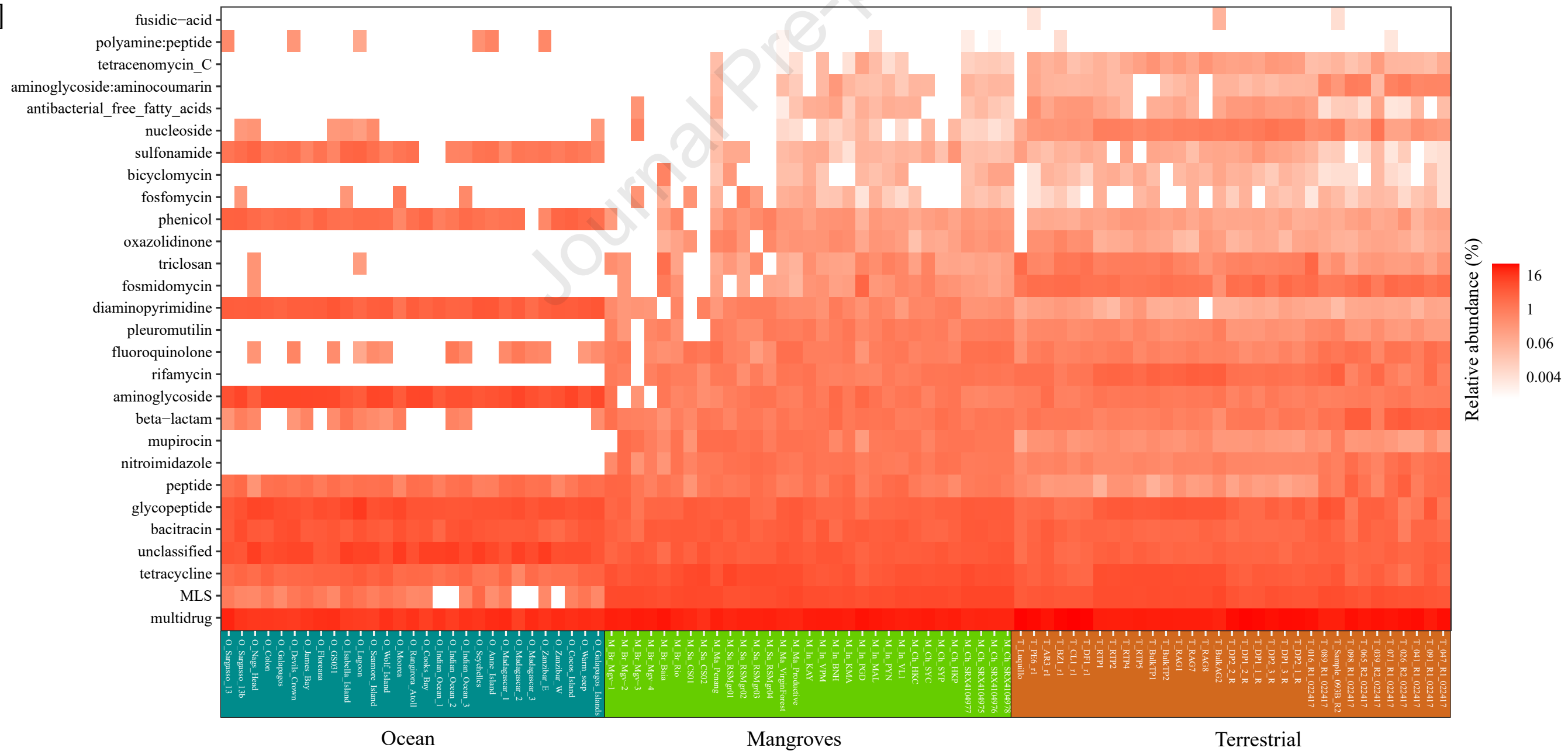




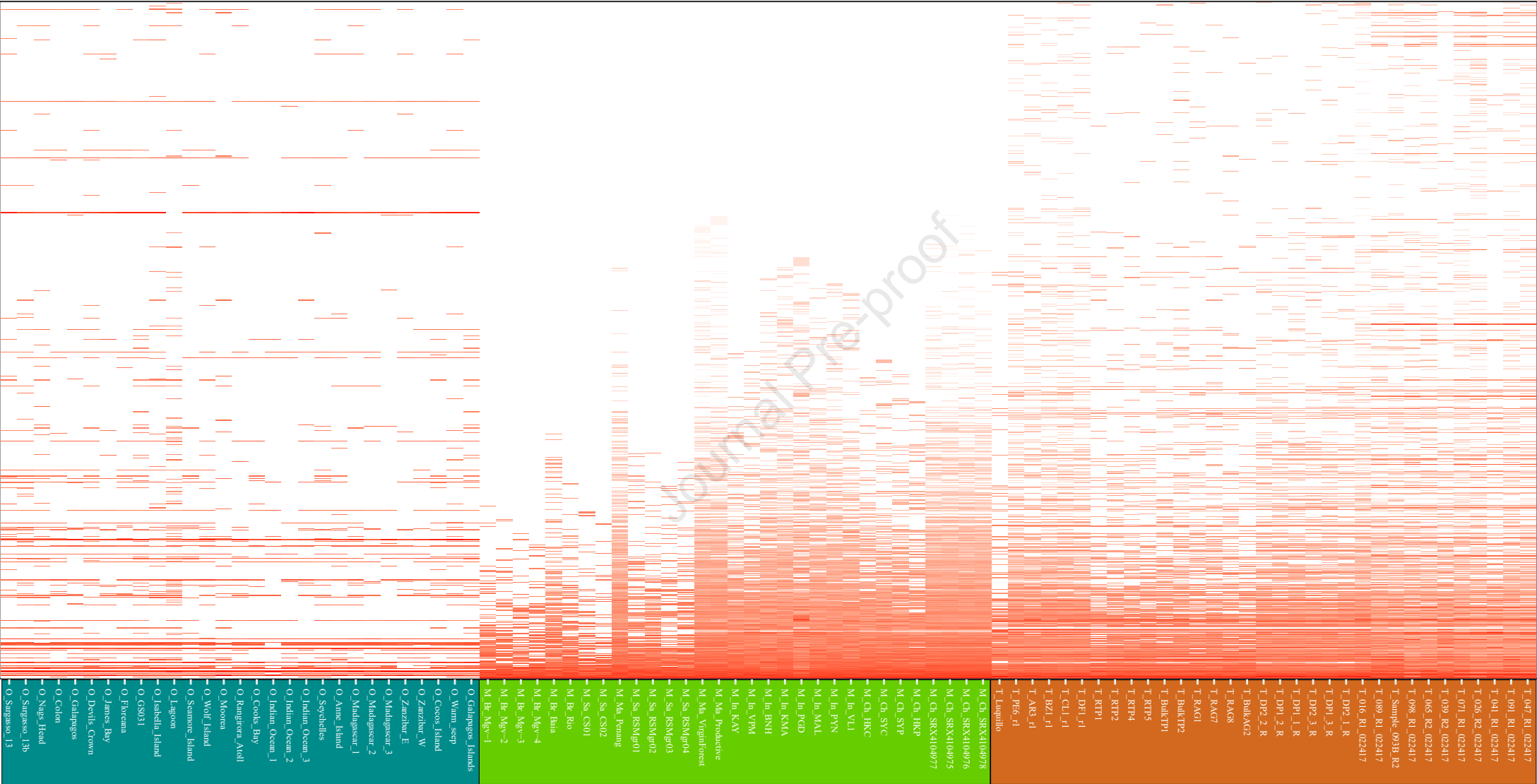
[A]

Ecosystems: ■ Mangroves ■ Ocean ■ Terrestrial

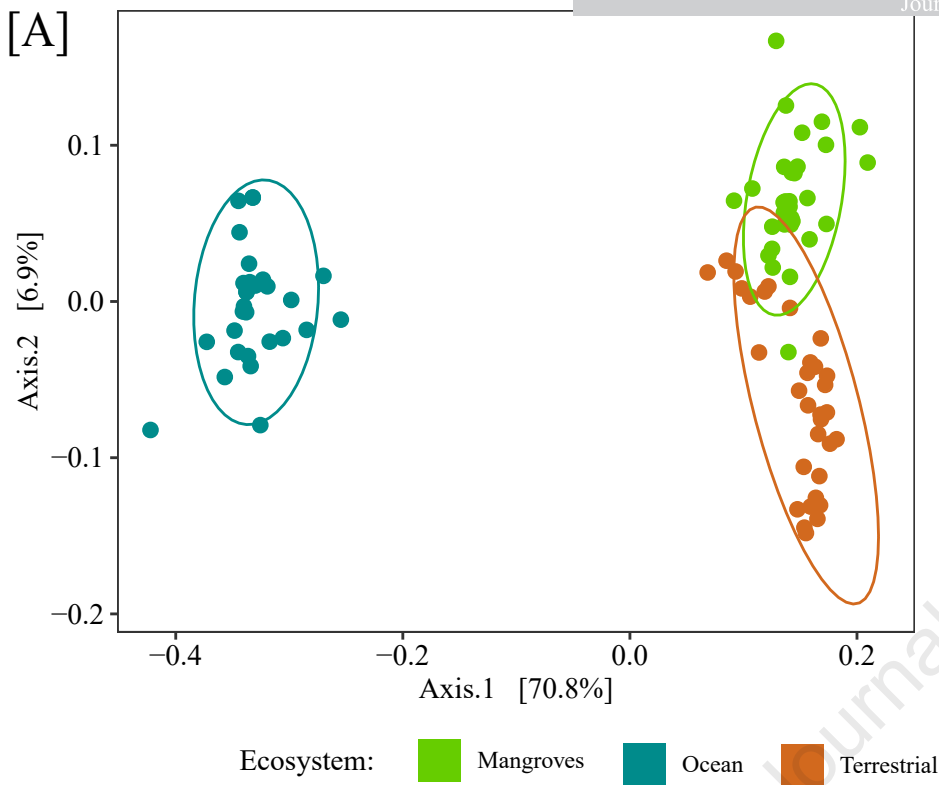
[D]



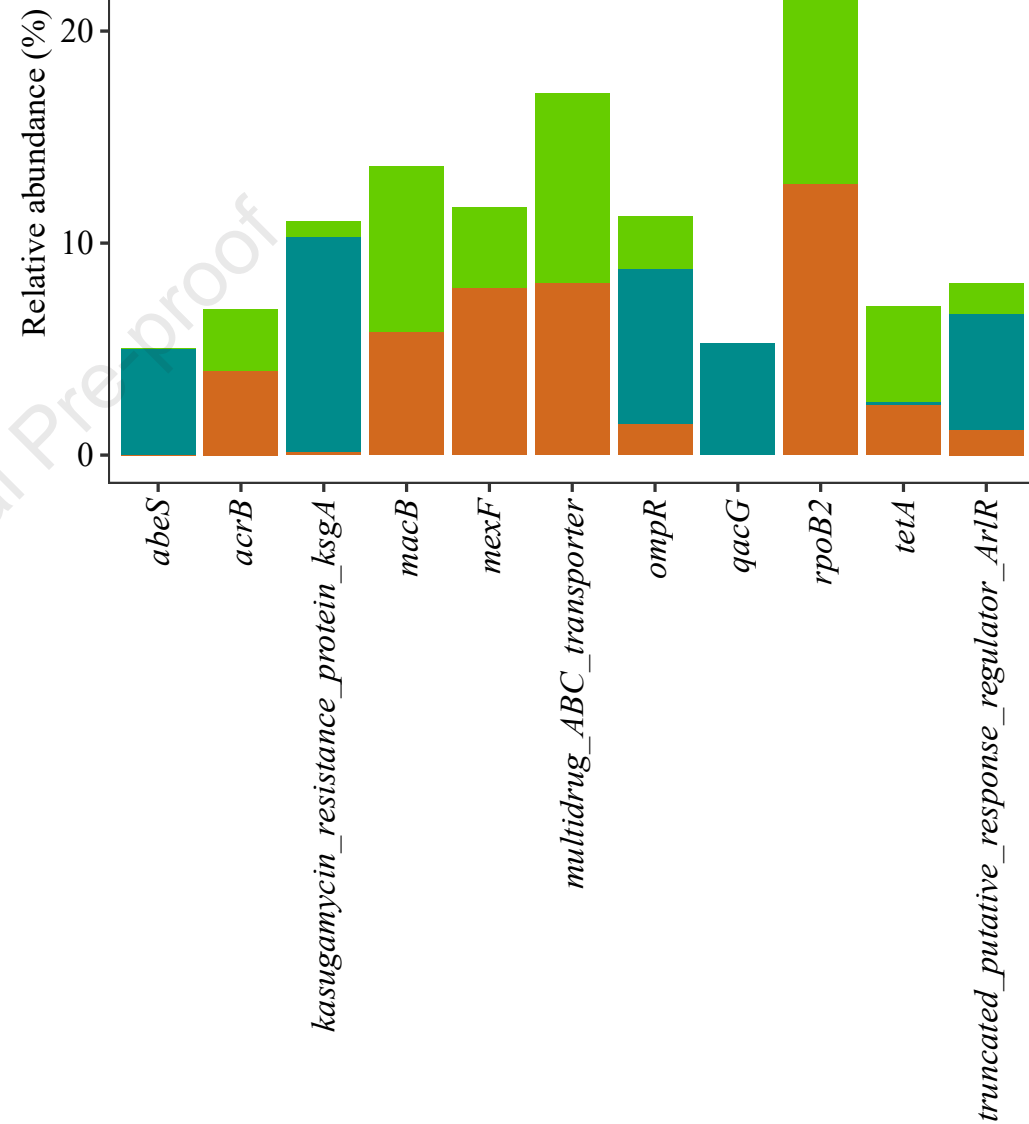
ARG Subtypes



[A]



[B]



Metagenomic insights into the antibiotic resistome of mangrove sediments and their association to socioeconomic status

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HIGHLIGHTS

- ARG diversity was correlated to human development index of the host country.
- Geography had higher influence on ARG diversity rather than pristine nature.
- Mangrove ecosystem was dominated by multidrug resistance.
- Mobile ARGs such as *tetX* and *mcr* were detected in mangrove metagenome.
- Ocean harbored the lowest ARG diversity followed by mangroves and terrestrial.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

N/A

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